

Goua Beedi

 Portfolio ·  GitHub ·  LinkedIn ·  gouabeedi@gmail.com ·  Paris, France

Profile

Final-year engineering student at ENSAI specializing in Statistics and Data Science, with experience in scoring, risk modeling, quantitative model evaluation, and decision-support systems. Particularly interested in statistical modeling, uncertainty analysis, and model reliability.

Skills

Programming: Python, SQL, R

Statistics & Quantitative Methods: probability theory, statistical inference, econometrics, Markov chains, survival analysis, optimization, calibration

Machine Learning: supervised and unsupervised learning, logistic regression, Random Forest, XGBoost, deep learning, model evaluation and robustness analysis

Generative AI & Information Retrieval: RAG systems, hybrid and vector search, evaluation of generative systems

Data Engineering: ETL pipelines, PostgreSQL, NoSQL, cloud computing, distributed data processing

Tools: FastAPI, Docker, Git, CI/CD

Experience

FINZUU – Agricultural Credit Scoring

2026

Data Scientist

- Designed an agricultural scoring engine to estimate production, income, agricultural risk, and farmers' repayment capacity.
- Developed scoring features and business rules from multi-country agricultural data, with explainable indicators and alerts integrated into a FastAPI service and a decision-support dashboard.

Square Management – Square Research Center

Apr. 2025 – Nov. 2025

AI & Data Science Intern

- Designed an experimental pipeline to compare more than 189 RAG configurations across different document types.
- Analyzed the impact of retrieval, chunking, indexing, and ranking strategies on response quality, robustness, and stability.
- Developed an evaluation system using LLM-as-a-judge to measure source fidelity and identify key failure modes.
- Implemented vector search with ChromaDB, exposed through a FastAPI service and containerized with Docker to support integration and deployment.

City of Paris – Mobility Agency

Jun. 2024 – Sep. 2024

Data Science Intern

- Developed a machine learning model to identify high-emission vehicle fleets from multi-source data.
- Designed a data pipeline combining more than 30k records with external APIs.
- Improved data quality and entity matching across heterogeneous data sources.
- Produced indicators and analyses to support operational and public-policy decisions.

Projects

Credit Scoring – Default Risk Assessment Tool (LCL Bank)

Dataset: 530k accounts, 143 variables

- Built a scoring pipeline including feature selection, data leakage checks, and stability tests.
- Compared logistic regression, Random Forest, and XGBoost to assess predictive performance, stability, and their ability to model default risk.
- Calibrated predictions and assessed model robustness to support decision-making.

Post-COVID Care Pathway Analysis – Georges-Pompidou European Hospital

- Analyzed longitudinal healthcare data to identify profiles of patients at risk.
- Used a Cox proportional hazards model to study rehospitalization risk in collaboration with hospital researchers.

Risk Assessment Decision-Support System (Zindi Competition)

- Designed a system combining machine learning scoring and business rules for risk assessment.
- Engineered features to improve robustness to incomplete data.
- Developed consistency checks and evaluation criteria to assess decision reliability.

Other projects: [beedigoua.github.io](https://github.com/beedigoua)

Education

ENSAI – Engineering Degree in Statistics and Data Science

2022 – 2026

Expected graduation: Dec. 2026. Selected coursework: probability theory, statistical inference, econometrics, Markov chains, survival analysis, optimization, statistical learning.

Preparatory Classes – Mathematics, Physics & Engineering Sciences

2020 – 2022

Languages

French (native) | English (professional working proficiency)